Median hours to merge pull request

Let's look at the median hours to merge pull requests where the pull request had at least one reviewer (Figure 4). There is better growth in open source where the median hours are around 11 hours. But you can see open source at work growing all the way up to 10 hours. With work, it is coming back to the pre-pandemic levels.

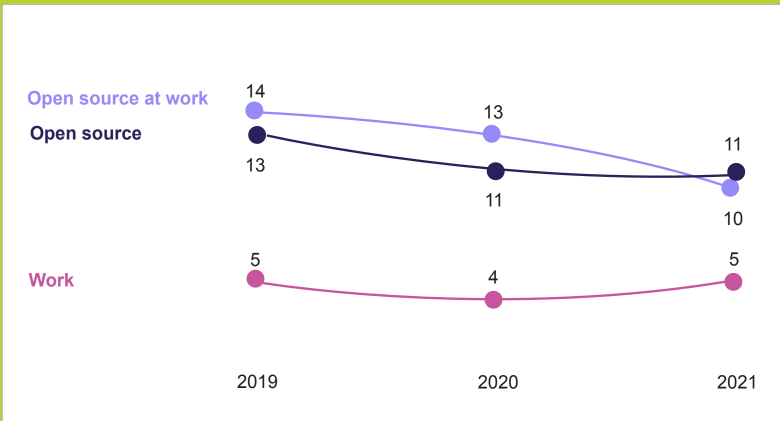

Figure 4: Pull requests with at least 1 reviewer

The stats in Figure 2 show the percentage of repositories with and without contributor guidelines. You may be wondering as to why a company should add contribution guidelines to its repositories. This is because these guidelines remove ambiguity and friction, and give a 17 per cent boost to productivity when developers know how they can contribute to repositories that belong to a different team or a community.

Open source repositories often use READMEs to share information and also to signal that the community is active. It is important to note that work repositories are used considerably less than open source projects, which may be because they share documentation and information in other internal forums.

GitHub issues are documentation too. Creating issues is the most common contribution. It invites non-coder contributors as well. This provides a great way for people to join, get help as well as manage projects. People can create issues within the first hour of joining GitHub. There is a way on GitHub itself to help you get started with this; it is called Good First Issues and is a great way to guide new members with their first contribution. Large repositories are most likely to use this. Repositories where almost 21 to 30 per cent of issues are not good generally have 13 per cent new contributors. You can see in Figure 3 the percentage of new contributors when a repository has Good First Issues and when it has no Good First Issues. This is 12 to 13 per cent more on average when you mark an issue as a Good First Issue.

Fast is better

Every developer, manager, and organisation wants tools and processes to be fast and easy. Similarly, open source project leaders and maintainers look for ways to make communities more welcoming. One of the ways to increase productivity is to improve the code and automate it, as it increases the overall speed of the work. It also improves collaboration and work culture. Good automation helps teams to communicate better and clearly, something that has been proven by many studies from time to time.

Developers' patterns on GitHub reflect that automating software delivery is a key enabler in open source and also helps teams to scale faster. Large repositories use features like GitHub Actions far more than small- or medium-sized repositories. When large repositories start using GitHub Actions, teams tend to merge two times more pull requests per day than before, which is almost a 61 per cent increase, and they merge 31 per cent faster. Repositories that are using GitHub Actions increase the number of merged pull requests. This increases the number of merged pull requests by 36 per cent while shrinking the merge time by 33 per cent.

You can improve your ability to merge pull requests quickly in the following ways:

- Assign no more than three reviewers to a pool request in an open source repository. This increases the chances that it will get merged within 24 hours.
- At work, pull requests with just one review are often merged within an 8-hour workday.
- Automation and community help us build better, together.

I hope these quick trends will help you understand what is really going on in the open source industry and where the contributors are coming from today. It is important to understand the relevance of documentation, work culture and the overall developer experience to repurpose your workflow accordingly. END

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